

Data & Planning: Model-Based & Offline RL

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- ▶ Day 7 (*The Reward Problem*) was about getting a *good signal* from the world: exploration (optimism, posterior sampling, curiosity) and designing or learning the reward
- ▶ One theme ran through it: what happens **off the data distribution**, where the agent has not been
- ▶ Today is about what to *do* with the data you have:
 - **Model-based RL**: learn a model of the world, plan with it, and fix mistakes by collecting more data where the model is wrong
 - **Offline RL**: learn from a *fixed* batch with no new interaction (the same off-distribution problem, with that fix *forbidden*)
- ▶ Two halves, one spine: distribution shift, and what you are allowed to do about it

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- No model
- Learn value function (and/or policy) from experience

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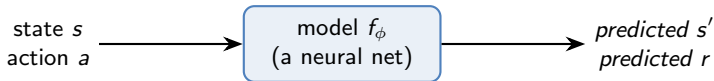
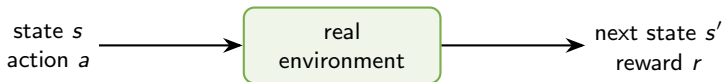
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► So far, we have only looked at model-free reinforcement learning

- ▶ A model is a **learned simulator of the environment**
- ▶ It is *not* a state, and *not* the real environment itself: it is a *function* (e.g. a neural network) that *imitates* how the environment behaves

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- ▶ It is *not* a state, and *not* the real environment itself: it is a *function* (e.g. a neural network) that *imitates* how the environment behaves
- ▶ You ask it: “*if I take action a in state s , what happens next?*”
- ▶ It answers with a predicted **next state** s' and **reward** r , just like the real environment would, but without having to actually be there

What is a Model?



the model *imitates* the environment

- ▶ Same inputs, same kind of outputs: the model is a stand-in we can query as often as we like, for free

What is a Model? (Formally)

- ▶ The environment is a Markov Decision Process $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R} \rangle$:
 - $\mathcal{P}$: the true transition dynamics (next-state rule), *unknown*
 - $\mathcal{R}$: the true reward function, *unknown*
- ▶ We assume the states $\mathcal{S}$ and actions $\mathcal{A}$ are known
- ▶ The model $\mathcal{M}_\phi = \langle \mathcal{P}_\phi, \mathcal{R}_\phi \rangle$ *learns* to approximate the unknown parts:
 $\mathcal{P}_\phi \approx \mathcal{P}$ and $\mathcal{R}_\phi \approx \mathcal{R}$
- ▶ Here ϕ are the model's **parameters**: the weights we train from data
- ▶ When the model predicts the next state directly we write $s' = f_\phi(s, a)$, the same f_ϕ from the diagram, used in the algorithm shortly

$$\underbrace{s_{t+1}}_{\text{next state}} \sim \underbrace{\mathcal{P}_\phi(s_{t+1} \mid s_t, a_t)}_{\text{predicted from current state \& action}}, \quad \underbrace{r_{t+1}}_{\text{reward}} \sim \mathcal{R}_\phi(r_{t+1} \mid s_t, a_t)$$

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- ▶ When the model predicts the next state directly we write $s' = f_\phi(s, a)$, the same f_ϕ from the diagram, used in the algorithm shortly
- ▶ We can also learn just *one* of the two (only dynamics, or only reward); often the dynamics $\mathcal{P}_\phi$ is the hard part

Why Build a Model of the Environment?

- ▶ A fair objection: if we can already *act* in the environment, why bother building an approximate *copy* of it?

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- ▶ You *cannot* ask “what would have happened if I had done something else?” (no what-ifs)
- ▶ You *cannot* reset it to any state you like
- ▶ You *cannot* look ahead without actually committing to the action
- ▶ Every single query costs a **real step**: time, money, wear, risk (a real robot can break; a real patient can be harmed)

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- ▶ **Sample efficiency**: turn a little real data into (almost) unlimited synthetic experience
- ▶ **Planning**: imagine the consequences of actions *before* committing to them
- ▶ Superpowers the real world denies you: ask counterfactuals, reset to any state, even differentiate through it
- ▶ It is trained by cheap **supervised learning**, and is somewhat task-agnostic (reusable across different rewards)

- ▶ No free lunch: the model is only an *approximation* of the real environment
- ▶ A *wrong* model can confidently mislead you. We will see exactly this failure (the “cliff”) shortly
- ▶ Model-based RL trades *real-world cost* for *model error*: worth it when the model is easier to get right than the policy itself

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- ▶ Goal: estimate the model $\mathcal{M}_\phi$ from logged experience $s_1, a_1, r_1, \dots, s_T$
- ▶ This is a **supervised learning** problem: each transition gives us an input (s, a) and the targets to predict (r and s')

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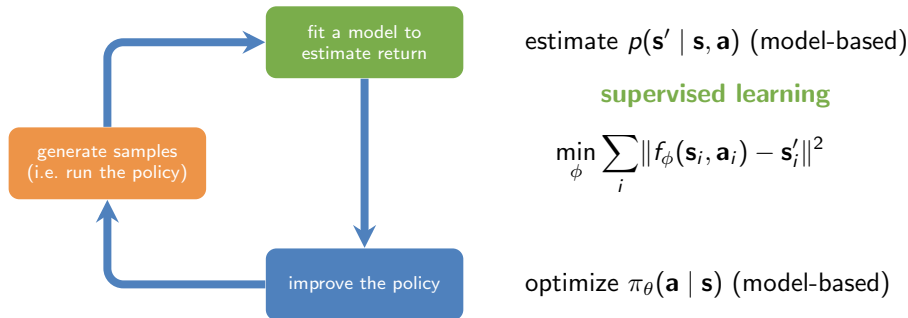
► Predict the next state: $s, a \rightarrow s'$

- the *same* action can lead to *different* next states: the environment is often random (stochastic)
- so a single guess is not enough; we want the whole *distribution* $p(s' | s, a)$ over where we might land
- learning a distribution is **density estimation**

- ▶ Each task comes with a matching loss:
 - regression (reward) $\rightarrow$ mean-squared error (MSE)
 - density estimation (next state) $\rightarrow$ maximum likelihood / KL divergence
- ▶ Common shortcut: if we *assume* the dynamics are deterministic (one action $\Rightarrow$ one next state), predicting s' collapses to plain regression too
- ▶ That is the f_ϕ trained with MSE, $\min_\phi \sum_i \|f_\phi(s_i, a_i) - s'_i\|^2$, that we use in the algorithm next

- ▶ We have the *what* (a learned simulator f_ϕ), the *why* (cheap, safe, plan-able experience), and *how to learn it*. Now: how to **use** it
- ▶ Two uses, and they organize the rest of today:
 - **Plan** with the model: pick good actions by looking ahead
 - **Optimize a policy** with the model: train a policy on simulated experience
- ▶ One perk we will lean on later: a model lets us *reason about its own uncertainty* (ensembles)

- Both uses run the same outer loop: collect data, fit the model, act better, repeat:



The bottom step (“improve the policy”) is exactly where the two uses *differ*: *plan* good actions directly (next), or *train* a policy π_{θ} on model-generated data (later).

- ▶ We take the first option (**planning**): use the model to choose good actions directly, with no separate policy yet. The simplest version:
 1. Run some policy (e.g. random policy) to collect data $D = \{(s, a, r, s')_i\}$
 2. Learn the model $f_\phi(s, a)$ (weights ϕ) to minimize $\sum_i \|f_\phi(s_i, a_i) - s'_i\|$
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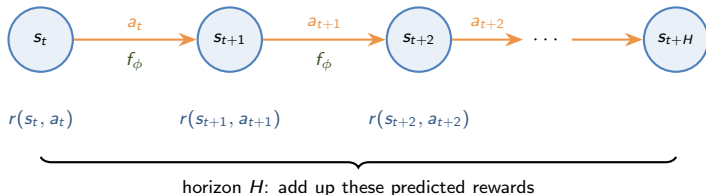
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- ▶ Step 3 is doing the real work, but *how* do we use the model to “choose actions”? Let us unpack it

How Does the Policy “Plan” with the Model?

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- ▶ Step 3 said to “*sample action sequences, run through the model to choose actions*”. What does that actually mean?
- ▶ **Planning**: imagine candidate action sequences, use the model to predict what each one leads to, and pick the one with the most reward
- ▶ No real environment needed: the model does the imagining. Let us build up the notation, one symbol at a time

- ▶ t : the **current** time step (where the agent is right now)
- ▶ H : the **planning horizon** (how many steps ahead we look)
- ▶ $a_{t:t+H}$: a whole **sequence** of actions ($a_t, a_{t+1}, \dots, a_{t+H}$); the “:” just means “from t through $t + H$ ”
- ▶ k : a counter for steps into the future, $k = 0, 1, \dots, H$, so s_{t+k} is “the state k steps from now”



- ▶ Start from now (s_t). Feed each state and a chosen action into the model f_ϕ to get the next state. This is “rolling out” the model H steps
- ▶ Each step earns a predicted reward $r(s_{t+k}, a_{t+k})$; we add them up

- Try this for many candidate sequences; keep the one whose roll-out scores highest:

$$a_{t:t+H}^* = \underbrace{\arg \max_{a_{t:t+H}}}_{\text{choose the sequence...}} \underbrace{\sum_{k=0}^H r(s_{t+k}, a_{t+k})}_{\text{... with the most predicted reward}}$$

rolling the model forward: $\underbrace{s_{t+k+1} = f_{\phi}(s_{t+k}, a_{t+k})}_{\text{each next state comes from the model}}$

- $a_{t:t+H}^*$ is the *best* sequence; we assume the reward $r(s, a)$ is known (or use the learned $\mathcal{R}_{\phi}$)
- The hard part is the arg max: searching over many sequences. We will see practical search methods shortly

- ▶ That is how a policy “plans through the model”: imagine action sequences, roll them forward with f_ϕ , and pick the best-scoring one
- ▶ This is exactly what step 3 does, so we now have a complete algorithm
- ▶ But will planning with a *learned* model actually work? Let us see how it can break. . .

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- ▶ The agent may learn that going right means that we can go higher!
- ▶ Thus, it will learn an incomplete policy and fall at the top.
- ▶ The model was only trained where the *random* policy went, but the planner now wants to go somewhere the model has never seen

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- ▶ π_f (the **final** policy after planning) wants to climb to the *top*
- ▶ So the states shift from the *data* distribution to the *learned-policy* distribution:

$$\underbrace{p_{\pi_0}(s)}_{\text{where we trained}} \neq \underbrace{p_{\pi_f}(s)}_{\text{where we now act}}$$

- ▶ The model is accurate on p_{π_0} but gets asked to predict on p_{π_f} , exactly where it never saw data
- ▶ Remember this picture; we return to it for *offline RL*

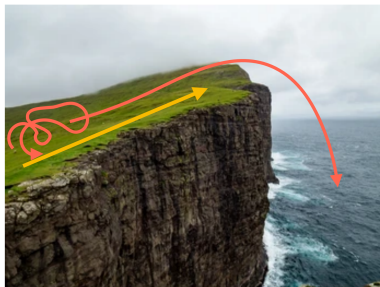
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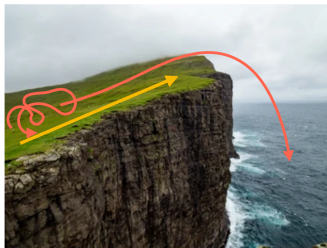
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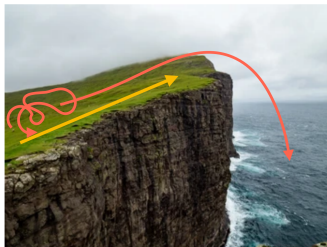
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- ▶ Now, we update the model using data collected with planning
- ▶ Then, replan periodically to help account for mistakes.

Revisiting the Cliff

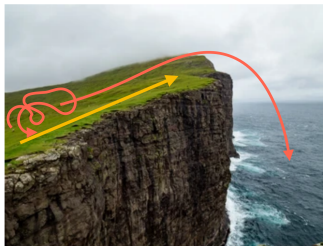




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- ▶ Then we re-train the model using the new observations
- ▶ The final policy is different now. It learns to go to the top and stop.

- ▶ Recall: planning = pick the action sequence whose model roll-out scores highest. The hard part was the $\arg \max$ over sequences
- ▶ The fix we just used (replan + collect data) also relied on this search
- ▶ Two separate questions: first, how to *search* for a plan (random shooting, CEM); then, how often to *replan* (MPC)

- ▶ Simplest idea (no gradients, just guess):
 1. Sample N action sequences $a_{t:t+H}^{(1)}, \dots, a_{t:t+H}^{(N)}$ from some distribution
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- ▶ Derivative-free and trivially parallel
- ▶ This is the planner in **Nagabandi et al.**: “random shooting”, no ensembles
- ▶ Weakness: most samples land in low-reward regions, wasteful

- Fix random shooting's waste: *refit* the sampling distribution toward the good sequences

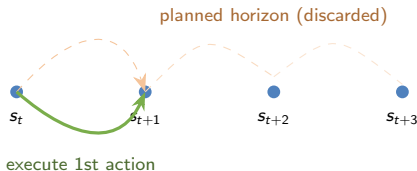
- ▶ Fix random shooting's waste: *refit* the sampling distribution toward the good sequences
 1. Sample N sequences from a Gaussian $\mathcal{N}(\mu, \Sigma)$
 2. Evaluate under the model; keep the top- k *elites*
 3. Refit μ, Σ to the elites
 4. Repeat for a few iterations
- ▶ The distribution concentrates on high-return plans
- ▶ This is the optimizer hiding inside step 3 of our algorithm

A Different Question: When to Replan?

- ▶ Random shooting and CEM answered *how to find* a good plan
- ▶ MPC answers a *different* question: not how to *search*, but **how much of a plan to trust before re-deciding**
- ▶ It is a planning *technique*, not another search method

- ▶ A full open-loop plan is fragile: model errors compound over the horizon H

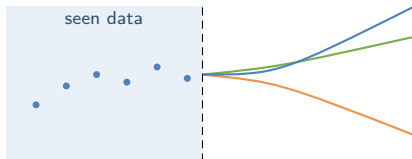
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- ▶ **MPC**: plan a horizon, but *execute only the first action*, then re-plan from the state you actually land in
- ▶ This *is* the “execute, then re-plan” loop we used to fix the cliff, now it has a name
- ▶ Next: combine a CEM planner with an *ensemble* model $\Rightarrow$ **PETS**

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- ▶ On data they *agree*; off data they *disagree*
- ▶ That disagreement measures **epistemic uncertainty** $u(s, a)$: a per state–action score for uncertainty due to *not having seen enough data*. It is high off-data, and it *shrinks* as you collect more (unlike the environment’s inherent noise, which never does)
- ▶ Plan conservatively where $u(s, a)$ is high: don’t trust a plan that exploits one model’s fantasy

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- ▶ Put the two pieces together:
 - **Planner:** CEM (from the previous section)
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- ▶ Evaluate each candidate plan by propagating it through *sampled* models and averaging the return
- ▶ **PETS** (Chua et al., 2018): this is the “CEM-based planner” in the case study; Nagabandi’s method is the same idea *without* the ensemble

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- ▶ We can also use model-based RL for policy optimization
- ▶ Key Idea:
 - Augment data with model-simulated roll-outs
 - This family (real data + model roll-outs) is **Dyna-style** (Sutton, 1991)

- ▶ We can also use model-based RL for policy optimization
- ▶ Key Idea:
 - Augment data with model-simulated roll-outs
- ▶ Algorithm (**MBPO**; Janner et al., 2019):
 1. Collect data using current policy π_θ , add to D_{env}
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- ▶ Compatible with variety of model-free RL methods
- ▶ MBPO's design: *short, branched* roll-outs from *real* states keep model error from compounding

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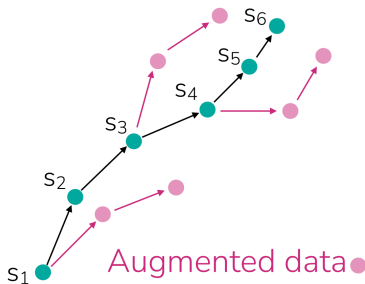
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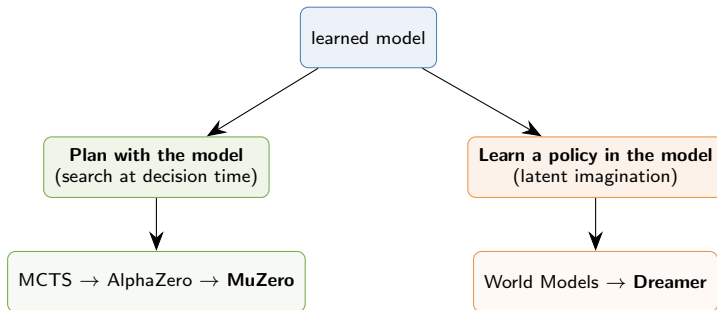
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Example real trajectory ●



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- ▶ Both *plan* by **tree search** (MCTS), the discrete-move cousin of our CEM planner: grow a tree of moves, search the promising branches
- ▶ **AlphaZero**: MCTS + a *known* model, the rules of Go / chess handed to it for free
- ▶ **MuZero**: *learns* the model instead, and the clever part is *what* it learns: a model that predicts only **reward, value, and the best move**, never the raw pixels (a model of *what matters*, not what the world looks like)
- ▶ So the same search works *without being told the rules*: one architecture mastered Go, chess, shogi, *and* Atari (Schrittwieser et al., 2020)

- ▶ The other family learns a policy entirely *inside* a learned world model: “imagined” roll-outs, no real interaction per update
- ▶ Its model is trained to **reconstruct what the world looks like** (predict the next observation), the opposite choice from MuZero’s “predict only what matters”
- ▶ **Dreamer / DreamerV3**: one fixed configuration solves 150+ tasks. Full treatment in the dedicated *World Models* lecture

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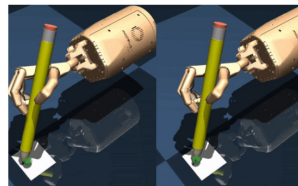
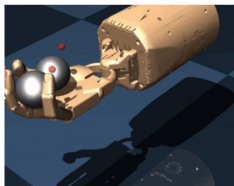
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Whether to use a model depends on how hard it is to learn!

Case study: Model-Based RL for dexterous manipulation



Tasks: Balancing balls (left) and handwriting (right) on a simulated dexterous hand

Model-free methods:

SAC: actor-critic method

NPG: policy gradient method

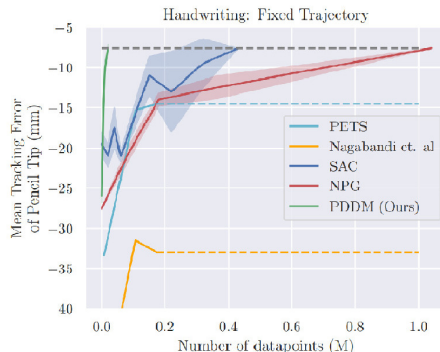
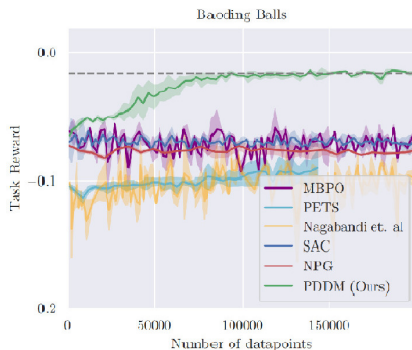
Model-based methods:

PDDM: proposed method

MBPO: RL with model-generated data

PETS: CEM-based planner

Nagabandi et al.: random shooting, no ensembles

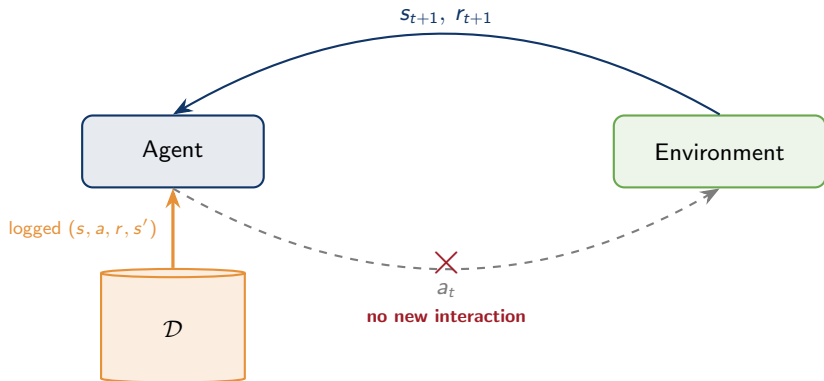


- Model-based **PDDM** (green) reaches higher reward and lower tracking error with *far less data* than the model-free baselines (SAC, NPG)

Nagabandi et al., *Deep Dynamics Models for Learning Dexterous Manipulation*

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- ▶ Every algorithm in this course has shared one quiet privilege: **if the agent acts badly, it can simply try again**
- ▶ DQN, PPO, DDPG, SAC, MBPO: all interleave *learning* with *acting*
- ▶ Even model-based RL fixed its early mistakes by going back out and collecting more data
- ▶ What if there is no “going back out”?



- ▶ Given: a **fixed** dataset $\mathcal{D} = \{(s_i, a_i, r_i, s'_i)\}$, collected by an unknown *behavior policy* π_β
- ▶ Goal: the best possible policy, **without a single new interaction**

- ▶ Healthcare: we cannot experiment on patients
- ▶ Autonomous driving: we cannot crash in order to learn
- ▶ Recommenders, industrial control: years of logs, no appetite for live exploration
- ▶ The situation of modern machine learning: **mountains of logged data, zero permission to explore**

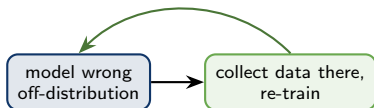


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- ▶ Part 1's fix for distribution mismatch was to *execute, observe, re-train*: go collect data where the model was wrong
- ▶ Offline RL is **exactly this problem, with the fix forbidden**

Model-based (part 1)



Offline (part 2)



- ▶ Careful, the failure *mechanisms* differ:
- ▶ The cliff was a **dynamics-model** error: f_ϕ predicted the wrong next state
- ▶ Offline RL's core failure is **value extrapolation**: Q invents values for actions the data never contains; it bites *even with perfect dynamics*
- ▶ Same disease (distribution shift), different organ (the model vs. the value function)

- ▶ Simplest possible idea: treat $\mathcal{D}$ as a supervised dataset and *imitate* it
- ▶ **Behavior cloning** (BC): train a policy π_θ to reproduce the data's actions:

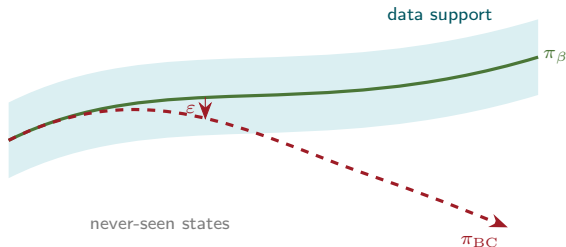
$$\min_{\theta} -\mathbb{E}_{(s,a)\sim\mathcal{D}} [\log \pi_\theta(a \mid s)]$$

- ▶ In words: at each state in $\mathcal{D}$, push π_θ to put *high probability on the action the data actually took*, i.e. $\max_{\theta} \mathbb{E}[\log \pi_\theta]$. Papers write this as “min of $-\log$ ” (the standard *negative log-likelihood loss*), since optimizers minimize losses

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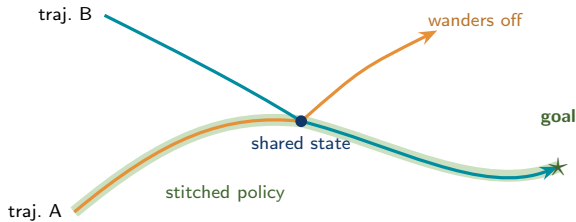
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- ▶ Plain supervised learning (like model learning, but for the policy): input s , target a , with no rewards, no Bellman, no bootstrapping. What could go wrong?



- ▶ π_β = the (unknown) policy that *collected* the data (β for “behavior”); π_{BC} = the clone we train
- ▶ Ceiling: at best π_{BC} *matches* π_β , copying its mistakes too
- ▶ Drift: each step the clone has a small chance ϵ of a wrong action $\rightarrow$ a slightly unfamiliar state $\rightarrow$ a bigger error $\rightarrow$ off the data entirely
- ▶ These errors *compound*: total error grows like $O(\epsilon T^2)$ over a horizon of T steps (Ross & Bagnell, 2010)

The Promise: Stitching



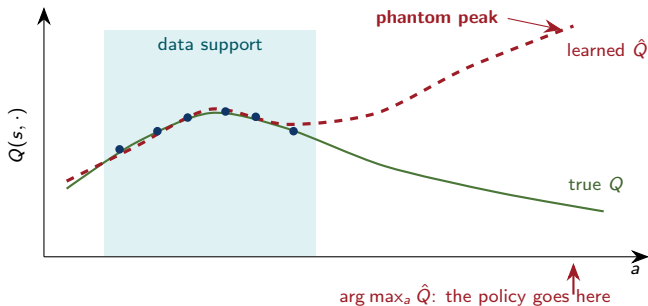
- ▶ $\mathcal{D}$ may contain **no good trajectory**, yet contain all the *pieces* of one
- ▶ Trajectories cross at a shared state: dynamic programming can **stitch** the good halves together
- ▶ The stitched policy is *better than anything in the data*; BC can never do this
- ▶ And we already own an off-policy algorithm: Q-learning learns from *any* data... right?

- ▶ Stitching needs **Q-learning**, so just run it on the fixed dataset. Look at its update target, which bootstraps from a max over actions:

$$y = r + \gamma \max_{a'} Q_{\theta}(s', a')$$

- ▶ That max ranges over **all** actions, including ones the dataset *never tried* (*out-of-distribution*, or **OOD**). Why is that fatal?

The Core Failure: Extrapolation Error

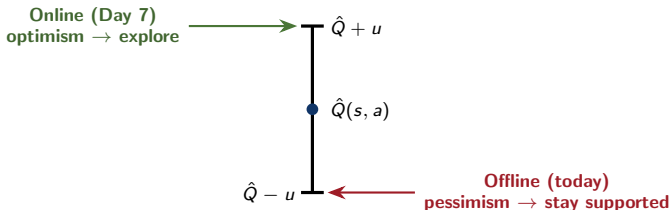


- ▶ Q_θ is a neural net: feed it *any* action and it returns a number, even where there is **no data**; there the value is not learned, only *extrapolated* (“invented”)
- ▶ The max seeks the *largest* invented value $\Rightarrow$ **overestimation**; the policy $\arg \max_a \hat{Q}$ chases that *phantom*, and bootstrapping spreads the inflated value onward

- Online, an overestimated action gets *tried*, disappoints, and is corrected; overestimation is self-erasing

- ▶ Online, an overestimated action gets *tried*, disappoints, and is corrected; overestimation is self-erasing
- ▶ Offline, the policy can chase the phantom forever: no data will ever contradict it
- ▶ More gradient steps make it **worse**, not better: we optimize against the fantasy (Fujimoto et al., 2019: *extrapolation error*)

Online, hype meets reality. Offline, it never does.



- ▶ Day 7's rule: *in the face of uncertainty, be optimistic*: UCB acts on $\hat{Q} + u$, because uncertainty was an **opportunity**
- ▶ Offline, uncertainty can never be resolved: it is a **hazard**
- ▶ Same machinery, opposite sign: act on $\hat{Q} - u$, *pessimism* in the face of uncertainty
- ▶ Two places to install it: in the **values** (CQL) or in the **policy** (BCQ, IQL)

- ▶ Train Q with an extra penalty that makes unfamiliar (OOD) actions *look bad on purpose*

$$\min_Q \alpha (\mathbb{E}_{a \sim \pi} [Q(s, a)] - \mathbb{E}_{a \sim \mathcal{D}} [Q(s, a)]) + \mathcal{L}_{\text{Bellman}}$$

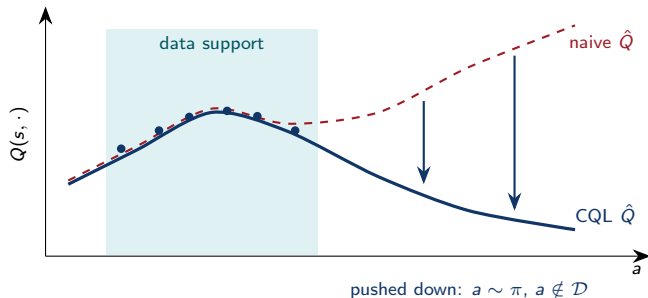
- ▶ $\min_Q =$ train Q (tune its weights) to make this whole expression small

- ▶ Train Q with an extra penalty that makes unfamiliar (OOD) actions *look bad on purpose*

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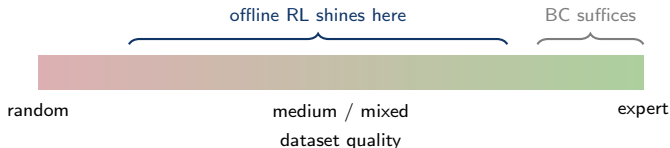
- ▶ $\min_Q$ = train Q to make this small. The penalty is a *difference*, so lowering it forces Q *down* on π 's (OOD) actions and *up* on the data's actions
- ▶ α = how hard to push ($\alpha=0$ is plain Q-learning); $\mathcal{L}_{\text{Bellman}}$ = the normal TD loss, fitting Q to $r + \gamma \max_{a'} Q(s', a')$

What CQL Does to the Phantom



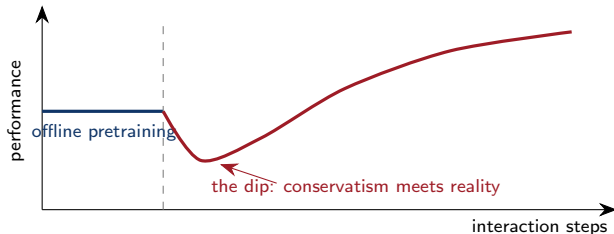
- ▶ The phantom peak from before, after CQL training
- ▶ OOD actions now look *worse* than in-data actions: the greedy policy stays home

- ▶ CQL enforced pessimism in the **values**. A second family enforces it in the **policy** instead:
- ▶ **Keep the policy close to the data**: let it pick only actions the behavior policy plausibly could have taken, so it never reaches an OOD action in the first place
- ▶ **BCQ** and **IQL** are the well-known instances; they differ in *how* they enforce this, and those details are genuinely involved (beyond our scope)
- ▶ Same principle as CQL, applied in a different place: *stay where the data is*



- ▶ Offline RL is measured on shared **benchmark suites** of fixed datasets: the standard one is **D4RL** (“Datasets for Deep Data-Driven RL”)
- ▶ Its tasks stress different skills: e.g. a maze-navigation set built to *require stitching*, or a dexterous-hand set of *human demonstrations*
- ▶ Each task comes at several *data qualities* (random / medium / expert mixtures), and method rankings flip between them
- ▶ Scores are normalized (0 = random, 100 = expert); on expert data plain BC is hard to beat, so **always run it as the baseline**

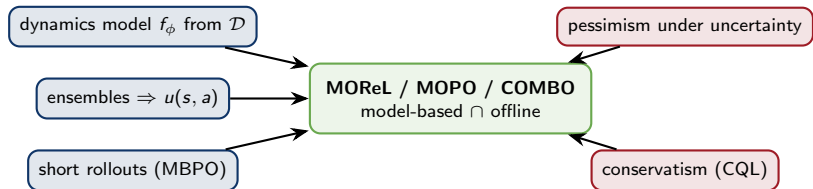
Offline → Online: The Deployment Recipe



- ▶ In practice: **pretrain offline** (data is free) → **fine-tune online** (interaction is scarce and expensive)
- ▶ Naive fine-tuning dips first: the pessimistic values get violently re-calibrated before they improve
- ▶ Fixes keep the offline ballast while admitting new data: AWAC, Cal-QL, RLPD
- ▶ You already know this recipe's most famous instance: *pretrain on logged human data, then fine-tune with RL*, the LLM pipeline (Day 9!)

Part 1: model-based

Part 2: offline



- ▶ Learn a model from $\mathcal{D} \rightarrow$ it is wrong off-support (the cliff!) $\rightarrow$ measure that with ensemble disagreement $\rightarrow$ build a *pessimistic MDP* $\rightarrow$ run MBPO inside it
- ▶ **MOREL, MOPO, COMBO** are all instances of *one* idea: learn a model from the batch, then *subtract a penalty wherever the model is uncertain* (e.g. reward $\hat{r} - \lambda u(s, a)$). This is pessimism, now about the *model*

- ▶ Exploration (Day 7) and offline RL (today) are the **same OOD problem, in reverse**
- ▶ Online: uncertainty is an opportunity, so add the bonus and go look
- ▶ Offline: uncertainty is a hazard, so subtract the bonus and stay home

Optimism for exploration, pessimism for offline data: the sign of the uncertainty bonus is the whole story.

- ▶ **Model-based RL**: learn the dynamics (and/or reward) by supervised learning, then *plan* (random shooting, CEM, MPC) or *optimize a policy* (Dyna-style, MBPO) with the model
- ▶ Models are wrong off-distribution: **ensembles** measure that, and re-collecting data (MPC) fixes it
- ▶ Modern lines: planning with a learned model (MuZero) and latent imagination (Dreamer)
- ▶ **Offline RL**: learn from a fixed batch, no interaction. The core failure is *value extrapolation* on OOD actions; the fix is **pessimism** (CQL, BCQ, IQL)
- ▶ The two halves meet in model-based offline RL (MOREL / MOPO / COMBO): a model plus a penalty for where it is uncertain
- ▶ One spine: *optimism* explores, *pessimism* stays safe; the sign of the uncertainty bonus is the whole story

Day 9: RL in the Real World (RLHF).

- ▶ RLHF's *reward model* is the learned reward model $\mathcal{R}_\phi$ from the start of today
- ▶ Learning a policy from fixed human demonstration / preference data is an **offline** setting
- ▶ “Pretrain on logged data, then fine-tune with RL” is exactly the offline-to-online recipe, now applied to LLMs

These slides have been adapted from

- ▶ Chelsea Finn & Karol Hausman, [Stanford CS224R: Deep Reinforcement Learning](#)
- ▶ Sergey Levine, [Berkeley CS285 Deep Reinforcement Learning](#)
- ▶ David Silver, Deepmind: [Introduction to Reinforcement Learning](#)

- ▶ Sutton (1991). *Dyna, an integrated architecture for learning, planning and reacting*.
- ▶ Chua et al. (2018). *Deep RL in a handful of trials (PETS)*. [arXiv:1805.12114](#)
- ▶ Nagabandi et al. (2019). *Deep dynamics models for learning dexterous manipulation (PDDM)*. [arXiv:1909.11652](#)
- ▶ Janner et al. (2019). *When to trust your model (MBPO)*. [arXiv:1906.08253](#)
- ▶ Silver et al. (2018). *A general RL algorithm that masters chess, shogi and Go through self-play (AlphaZero)*. *Science* 362(6419).
- ▶ Schrittwieser et al. (2020). *Mastering Atari, Go, chess and shogi by planning with a learned model (MuZero)*. [arXiv:1911.08265](#)
- ▶ Hafner et al. (2023). *Mastering diverse domains through world models (DreamerV3)*. [arXiv:2301.04104](#)

- ▶ Ross & Bagnell (2010). *Efficient reductions for imitation learning* (compounding error).
- ▶ Fujimoto et al. (2019). *Off-policy deep RL without exploration* (BCQ; extrapolation error). [arXiv:1812.02900](#)
- ▶ Kumar et al. (2020). *Conservative Q-learning for offline RL* (CQL). [arXiv:2006.04779](#)
- ▶ Kidambi et al. (2020) *MOReL*; Yu et al. (2020) *MOPO*; Yu et al. (2021) *COMBO*.
- ▶ Kostrikov et al. (2021). *Offline RL with implicit Q-learning* (IQL). [arXiv:2110.06169](#)
- ▶ Fu et al. (2020). *D4RL: datasets for deep data-driven RL*. [arXiv:2004.07219](#)
- ▶ Offline→online fine-tuning: Nair et al. (2020) *AWAC*; Nakamoto et al. (2023) *Cal-QL*; Ball et al. (2023) *RLPD*.